

HILLOL DAS PIAS

AI-Native Full-Stack Engineer · 4.5+ Years Experience
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PROFESSIONAL SUMMARY

AI-native full-stack engineer with 4.5+ years building production web applications and a growing specialisation in LLM-powered product development. Hands-on with the OpenAI and Google Gemini APIs, LangChain, LangGraph, RAG pipelines (ChromaDB), tool-calling AI agents, and multi-agent systems — including a deployed ReAct agent orchestrating 17 tools with LangSmith tracing, a test-gated CI/CD pipeline, and Docker-based deployment. Proficient across the modern JavaScript ecosystem (Next.js, Node.js, Express.js, React, TypeScript) and Python backends (FastAPI, async SQLAlchemy). Proven delivery across RESTful APIs, real-time systems (Socket.IO), headless CMS architectures, e-commerce platforms, and containerised services via Docker and CI/CD pipelines. Collaborative and self-driven, with experience across the full development lifecycle in remote-first environments.

CORE COMPETENCIES

AI / LLM LangChain · LangGraph · RAG pipelines (ChromaDB) · Multi-agent & ReAct agents · OpenAI & Google Gemini APIs · Tool use & function calling · LLM response streaming · LangSmith tracing · Prompt engineering

Full-Stack Next.js · React · Node.js/Express · TypeScript · FastAPI · Socket.IO · REST APIs · Java (Spring Boot)

Infra & Data Docker · Docker Compose · GitHub Actions CI/CD · PostgreSQL · MySQL · Neon · async SQLAlchemy 2.0 · Alembic · MongoDB · ChromaDB (vector DB) · Vercel · Render · Railway · Stripe · Strapi CMS · Supabase

Testing & Quality pytest (async) · CI-gated deployments · dependency-injected test isolation · Alembic migration review

System Design (working knowledge): Redis caching, load balancing, CDNs, message queues, read replicas, object storage, and AWS (ECS/RDS) — architectural understanding applied in design discussions and interview prep; not yet deployed in a production project.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | SwaPE IT Solutions

Dec 2023 – Present

- Architected and shipped production web apps using Next.js (App Router), Node.js, React, and Python (FastAPI) services, with fully responsive UIs via Tailwind CSS and Framer Motion.
- Led system design decisions and defined scalable architecture patterns across multiple client projects.
- Integrated LLM APIs (OpenAI, Google Gemini) into client-facing features, gaining hands-on experience with prompt engineering and AI-powered product flows.
- Improved code quality through automated tooling, linting, peer reviews, and performance profiling.
- Integrated third-party services and APIs, reducing development time by establishing reusable integration patterns.

Full-Stack Developer | Raj IT Solutions Ltd.

Nov 2021 – Dec 2023

- Built RESTful APIs with Next.js, Node.js (Express), and Python (FastAPI); implemented real-time features using Socket.IO serving concurrent users without performance degradation.
- Developed and maintained backend microservices (including Java/Spring Boot) ensuring high availability and horizontal scalability.
- Integrated Strapi headless CMS, enabling non-technical stakeholders to manage content without developer intervention.
- Established CI/CD pipelines with Docker, cutting deployment time significantly and reducing production incidents.

Freelance Full-Stack Developer | Independent

Oct 2015 – Present

- Building and shipping AI-agent and AI-assisted products (see AI Projects below), spanning Next.js frontends, FastAPI/Node backends, and LangChain/LangGraph agent orchestration.
- Delivered e-commerce and booking systems using Node.js, React, and Next.js with Stripe payment integration, role-based access control, and RESTful APIs.
- Designed and optimised PostgreSQL, MongoDB, and MySQL databases for scalable, production-ready applications.

Graphic Designer | Cloudville

Jan 2020 – Oct 2021

- Designed digital assets and social media content; developed strong visual communication and client-facing skills directly applicable to frontend UI work.

AI PROJECTS

Support Ops Copilot — Multi-Agent Customer Support System (Building) · *LangGraph · FastAPI · OpenAI API · ChromaDB · MCP*

- Agent orchestration & RAG: LangGraph pipeline (triage → retrieval → draft → guardrail → approval → execute) grounding responses in a live knowledge base via ChromaDB.
- Guardrails & security: LLM-as-judge faithfulness scoring combined with rule-based checks for prompt injection and credential leakage.
- Human-in-the-loop control: LangGraph interrupt() genuinely pauses execution for approval before any customer-facing action is taken.
- MCP tool integration: customer actions defined as MCP-shaped tool schemas, ready to connect to real Gmail/Slack servers.
- Cost & latency observability (in progress): per-node token and latency tracking toward a cost-per-resolution dashboard.
- Evals (in progress): golden-ticket suite scoring accuracy and faithfulness, gating CI on quality thresholds.

Ledger — AI-Powered Task & Project Manager · *FastAPI · Next.js · PostgreSQL · LangGraph · ChromaDB · LangSmith · Docker*

- Built a ReAct-style AI agent (LangGraph) with 17 tools for CRUD operations and RAG-based semantic search (ChromaDB), streaming responses token-by-token; traced end-to-end in LangSmith.
- Designed async REST API (FastAPI, SQLAlchemy 2.0) with Alembic migrations, deployed on PostgreSQL (Neon).
- Built a full CI/CD pipeline (GitHub Actions): an automated pytest suite against an ephemeral Postgres instance gates both Docker image builds (Docker Hub) and production deploys (Render).
- Diagnosed and resolved LLM tool-selection ambiguity and argument-mapping errors in the agent's tool-calling logic using LangSmith traces — failures invisible to standard error handling, since the agent returned confident but incorrect answers rather than throwing exceptions.
- Containerised with Docker and Docker Compose (healthchecks, named volumes, service networking) for reproducible local development.
- Built a custom-designed Next.js/TypeScript frontend with a live-streaming chat UI.

Live demo: taskapi-frontend.vercel.app · Frontend: github.com/HilloIPias/taskapi-frontend · Backend: github.com/HilloIPias/taskapi-backend

Cookmind — AI Recipe Generator · *Next.js · TypeScript · Google Gemini API · Strapi · PostgreSQL · Clerk · Arcjet*

- Built an AI recipe generator that analyses a photo of a user's ingredients via the Google Generative AI (Gemini) API and returns tailored, ready-to-cook recipes, with drag-and-drop image upload and PDF export of results.
- Implemented Clerk-based authentication and Arcjet bot/rate-limit protection on a Next.js 16 (App Router) + React 19 frontend, backed by a Strapi 5 headless CMS on PostgreSQL.

Live demo: cookmind-ivory.vercel.app · GitHub: github.com/HilloIPias/cookmind

ADDITIONAL PROJECTS

- Chatty App — real-time full-stack chat application (JavaScript). GitHub
- Product Store (MERN) — full-stack e-commerce product catalog. GitHub
- NextLevel Food/Foodies — recipe sharing application. GitHub

EDUCATION

BSc, Applied Physics & Electronic Engineering — University of Rajshahi

COURSES & CERTIFICATIONS

Agentic AI Engineering with LangChain & LangGraph (Udemy) · Functions, Tools & Agents with LangChain (DeepLearning.AI) · React — incl. Next.js, Redux (Udemy) · FastAPI Complete Course (Udemy) · Full-Stack Web Dev Bootcamp (Udemy) · CCNA — RUET